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Adaptive real-time electronic-nose system for multi-gas monitoring

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A B S T R A C T

Electronic-nose systems often suffer from cross-response interference, sensor drift, and high maintenance costs under complex operating conditions. To address these challenges, a mixed-gas concentration prediction system with real-time processing and intelligent self-calibration is developed. A dynamic signal-based phase identification strategy enables automated end-to-end processing from raw sensor signals to concentration prediction. By integrating a contractive autoencoder with a deep supervision network (CA-DSN), the model enhances feature robustness and improves quantitative accuracy in mixed-gas environments. An adaptive calibration mechanism selectively performs lightweight fine-tuning or full retraining based on real-time prediction errors, reducing training cost by approximately 60% without measurable performance degradation. For $C_2H_5OH/C_7H_8/NH_3$ mixtures, the average mean absolute error (MAE) is 1.33 ppm, while for $H_2/CO/C_2H_2$ mixtures, it decreases to 0.69 ppm. The system exhibits consistently strong performance across chemically distinct gas systems, demonstrating its cross-system generalizability. In addition, the system incorporates an adaptive calibration mechanism that helps reduce the maintenance burden associated with sensor performance variations. This work offers a viable pathway toward robust, low-maintenance electronic-nose platforms suitable for long-term deployment.

1. Introduction

Gas sensors are widely used in industrial and daily applications, playing key roles in food safety [1,2], pollution control [3], medical diagnosis [4] and early detection of battery thermal runaway [5]. With the rapid development of the Internet of Things (IoTs), gas sensors are transitioning from basic concentration alarms to intelligent analytical devices [6–11]. Thus, integrating sensor arrays with artificial intelligence algorithms is critical for real-time detection in complex gas environments.

Conventional gas analysis techniques, such as gas chromatography–mass spectrometry (GC–MS) [12], Fourier-transform infrared spectroscopy (FTIR) [13] and ultraviolet–visible absorption spectroscopy [14], rely on the physical and chemical properties of gases to achieve component and concentration analyses through molecular separation or spectral absorption. However, these methods are hindered by complex procedures, high costs and long detection time. In contrast, electronic-nose systems offer a promising alternative, providing high selectivity, precision, rapid response, stability and scalability [15,16]. For instance, Wang *et al.* [17] applied spiking neural networks (SNNs) to classify four gases, achieving a classification accuracy of 100%. Chen *et al.* [18] employed a convolutional neural network (CNN) with mixed convolutional kernels to classify six gases with an accuracy of 99.72%. Li *et al.* [19] further developed a hardware-based recognition system that integrates optically excited sensors with a memristor array, enabling precise classification of four gases with 100% accuracy. These studies highlight the strong potential of deep learning–based approaches to substantially enhance the performance of electronic-nose systems.

Despite the promising potential of electronic-nose technology, its long-term deployment in complex real-world scenarios remains challenging. Most existing studies are conducted under single-gas conditions and therefore fail to capture the complexity and interdependence of gas mixtures encountered in practical environments [20]. As a result, such approaches are difficult to generalize to real-world settings with dynamically varying multi-component gas compositions, making the accurate identification and prediction of mixed-gas concentrations a demanding yet highly meaningful research challenge.

Recent advances have explored deep learning–based solutions for mixed-gas sensing. Jia *et al.* [21]. proposed an AGT-TCN framework that integrates a gated recurrent unit (GRU) variant with temporal convolutional networks (TCNs), enabling effective extraction and learning of both long-term and short-term temporal features. Tang *et al.* [22]

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developed a Mamba-based model for mixed-gas concentration prediction, achieving MAEs of 1.4、2.2 and 2.5 ppm for ternary mixtures of H_2, CO, and C_2H_4. Gan *et al.* [23] utilized variational mode decomposition (VMD) to extract informative features from a small-scale dataset for predicting the concentrations of CO and C_2H_4, yielding MAEs of 1.33 ppm and 2.02 ppm, respectively. In addition, Ma *et al.* [24] further proposed a domain difference maximum entropy (DDME) technique to facilitate model calibration under varying conditions.

Nevertheless, most deep learning–based approaches rely on pre-collected, large-scale static datasets, which require extensive preprocessing and consequently cannot support real-time processing of continuously generated sensor signals [25–27]. In addition, environmental variations and sensor baseline drift present persistent challenges to long-term deployment. Conventional strategies, such as periodic manual recalibration [28], often entail system downtime and substantial maintenance costs, limiting their suitability for unattended operation. These challenges highlight the need for electronic-nose systems capable of real-time processing of continuous sensor data and adaptive response to changing environmental conditions without frequent manual intervention.

To address these limitations, this study proposes a mixed-gas detection approach based on a CA-DSN model coupled with a real-time training and calibration mechanism. The framework integrates dynamic signal-based phase identification with adaptive model updating, enabling an end-to-end learning and inference pipeline for mixed-gas concentration prediction. The key contributions are as follows:

1. **Real-time phase identification strategy** enabling automated, end-to-end processing from sensor signals to gas concentration prediction.
2. **Ultra-lightweight model** capable of accurate prediction for three-component gas mixtures.
3. **Adaptive self-calibration mechanism** enabling model adaptation to sensor performance variations and reduced maintenance requirements.
4. **Gaussian process regression (GPR) and particle swarm optimization (PSO)** reducing the number of labeled training samples required.

Experimental results show that the system achieves ppm-level accuracy across two ternary and three binary gas mixtures, as well as under different temperature and humidity conditions, indicating consistent performance under varying environmental settings. The proposed adaptive calibration strategy reduces training costs by approximately 60% with no noticeable performance degradation in experiments. Furthermore, satisfactory initial model training can be achieved using only 40 distinct samples, suggesting that the proposed framework can support efficient model development under limited data conditions.

2. Materials and methods

To enable real-time and accurate detection of mixed-gas concentrations, a complete electronic-nose system was developed, as illustrated in Fig. 1. The system integrates a gas delivery platform, sensor array, data acquisition module, and core algorithmic components into a unified end-to-end framework. This section describes the hardware configuration, gas sample design, and data acquisition procedures that support the operation of the proposed system.

2.1. Experimental setup

The dataset was collected using the gas-testing platform shown in Fig. 2. All tests in this study were conducted at 25 ℃. C_2H_5OH at 150 ppm, NH_3 and C_7H_8 at 100 ppm from the Shenkai Gas were used, with air as the background and dilution gas. Mass flow controllers (MFCs) precisely regulated the gas proportions, creating mixed-gas environments with varying concentrations and compositions.

The sensor array comprised five different models (MP702, TGS2600, TGS2602, TGS2603 and TGS2610) from Winsen and Figaro, all integrated within the test chamber to ensure consistent gas exposure. Sensor resistance was collected at a 1 Hz sampling rate using custom signal conditioning circuits and a data acquisition card, with data transmitted to a computer for real-time processing, feature extraction and model computation. All sensors operated at a constant 5 V. Prior to testing, aged sensors were exposed to air for 5000 s to establish a stable baseline. Each gas sample was introduced for 200 s, followed by a 400 s recovery period, with mass flow controllers (MFCs) regulating the gas flow according to the following equation:

$$S_{test} = \frac{C_{test}}{C_{origin}} \times S_{total} \tag{1}$$

Here, S_{test} represents the flow rate of the target gas, C_{test} and C_{origin} to

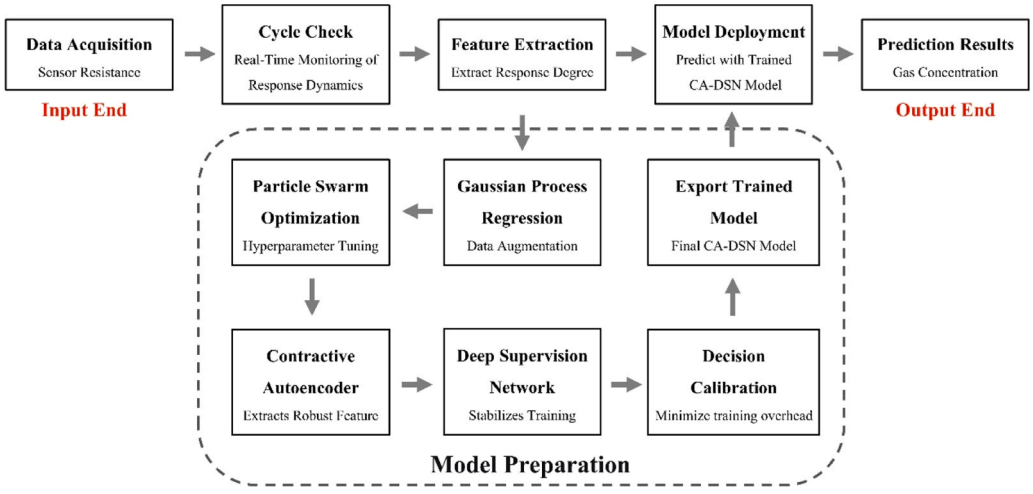


Fig. 1. Overall workflow of the proposed intelligent gas sensing system. The pipeline begins with raw sensor resistance acquisition, followed by real-time cycle checking based on response dynamics and automated feature extraction. The extracted features are fed into the deployed CA-DSN model to generate gas concentration predictions at the output end. The lower panel illustrates the model preparation process: a Contractive Autoencoder is used for robust feature extraction, while a Deep Supervision Network stabilizes training. Gaussian Process Regression augments the dataset, and Particle Swarm Optimization performs hyperparameter tuning. Decision calibration further minimizes training overhead, after which the optimized CA-DSN model is exported for deployment.

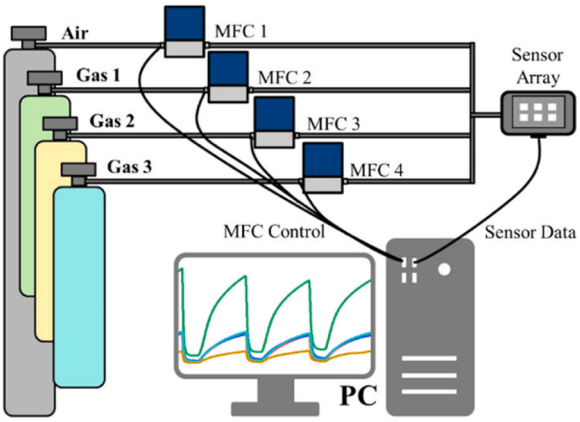


Fig. 2. Schematic of the gas-sensing test platform. MFCs precisely regulate the flow of synthetic air (background/dilution gas) and three target gases to create mixed-gas environments with varying concentrations and compositions. The sensor array, housed within a test chamber, is exposed to the gas mixtures. Resistance data is acquired and transmitted to a PC for processing, feature extraction, and model computation.

the target and original concentrations of the gas, respectively, and S_{total} represents the total gas flow rate. The sensor response (a) during the measurement is defined as follows:

$$a = \frac{R_g}{R_0} \tag{2}$$

Here, R_g represents the sensor resistance upon exposure to the test gas and R_0 denotes the baseline resistance.

2.2. Experimental design

To evaluate system performance, a concentration dataset for a ternary gas mixture was designed and collected. C_2H_5OH, NH_3, and C_7H_8—common VOCs that may co-occur in the same environment—were selected, because these gases exhibit highly similar sensor response trends, substantial overlap in response dynamics, and severe cross-interference, thereby providing a particularly challenging scenario for accurate identification. To further validate the model's generalizability, additional datasets were included, comprising three binary VOC mixtures, and ternary mixtures (H_2, CO and C_2H_2) collected under different humidity and temperature conditions.

To further approximate real-world gas environments, each gas was prepared at 20 concentration levels: The concentrations of C_2H_5OH were tested from 2.25 to 45 ppm in 2.25 ppm increments, while NH_3 and C_7H_8 were tested from 1.5 to 30 ppm in 1.5 ppm increments. Using a constrained random sampling method under dry conditions, 125 ternary mixed-gas samples were generated, covering a wide concentration range and ensuring a diverse yet uniform distribution of mixture proportions. This experimental design enhances the model's robustness and generalizability. The fine-grained concentration increments compel the model to capture subtle nonlinear variations in sensor responses at low ppm levels, while the constrained random sampling generates more complex gas combinations, partially simulating the distribution of gases in real-world environments. Some of sensor response curves for the samples are shown in Fig. 3.

Of the 125 samples, 120 were used for model training, validation, and comparative performance evaluation. The remaining 5 samples, representing continuously unknown data, were reserved to simulate real deployment conditions and assess real-time processing performance. To simplify the model comparison process, an equivalent processing scheme was applied during model training to emulate the system's real-time workflow, with details provided in Section S1. Information on the composition of the ternary mixture dataset, experimental procedures, and sample acquisition methods is provided in Section S2 and S3.

Unit : ppm	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5	Sample 6
C_2H_5OH	42.75	38.25	4.5	11.25	24.75	9
NH_3	9	16.5	6	28.5	4.5	27
C_7H_8	25.5	15	27	19.5	4.5	12

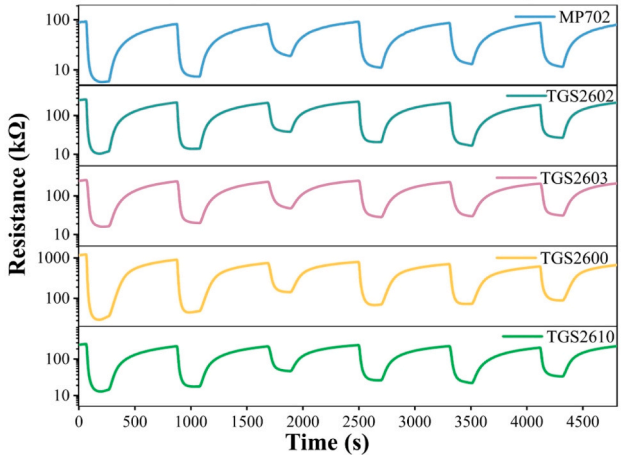


Fig. 3. Sensor response curves for C_2H_5OH, NH_3, and C_7H_8. The table inset lists the concentration (in ppm) of each gas component in the corresponding samples. The curves illustrate the dynamic resistance changes during the 200-second gas exposure and subsequent 400-second recovery period in air.

All algorithms were executed on a Windows 10 workstation with an AMD Ryzen 7 7700 CPU, NVIDIA GeForce RTX 4080 Super GPU, and 32 GB of memory. Experiments were conducted in Python 3.12 using the PyTorch 2.1.1 deep learning framework and CUDA 11.8 for network model construction. All model evaluation results were obtained using 10-fold cross-validation.

3. Methodology

This study aims to achieve fully automated, efficient, and intelligent end-to-end real-time processing from sensor response signals to concentration prediction, with minimal human intervention and without preprocessing or curve-smoothing steps. Key techniques employed are summarized as follows. First, a real-time phase identification strategy based on the first derivative of sensor responses was implemented, enabling signal acquisition, feature extraction, and subsequent processing to operate as a fully automated pipeline. Gaussian process regression (GPR) [29] was applied for data augmentation to expand the training dataset while minimizing the need for labeled samples. A CA-DSN model, integrating a CAE [30] with a DSN [31], was employed for concentration prediction, enhancing robustness, generalization, and accuracy in complex gas environments. Model parameters were optimized using particle swarm optimization (PSO) to further improve predictive performance. Finally, an efficient online calibration and training mechanism was incorporated, allowing real-time prediction errors to guide lightweight parameter refinement or full retraining, thereby maintaining long-term reliability and compensating for sensor baseline drift. The overall system workflow and operational logic are presented in the following sections.

3.1. Dynamic, derivative-guided real-time feature extraction

To overcome the limitations of traditional feature extraction methods that rely on fixed time windows or human intervention, a dynamic decision mechanism based on the first-order derivative of the sensor response was implemented, enabling real-time extraction of raw signals. During real-time monitoring of sensor resistance, the first-order derivative of the signal curve ($\frac{dR}{dt}$) is continuously calculated. This de-

rivative captures the overall evolution trend of the sensor response rather than relying on experimental timing, and is used to determine the current response stage. Therefore, the segmentation process is driven by the temporal evolution of the signal and is not constrained by the specific acquisition protocol.

This derivative captures the evolving trend of the sensor response and is used to determine the current response stage of the sensor, enabling segmentation driven by intrinsic signal dynamics rather than experimental timing.

As illustrated in Fig. 4, Stage ① corresponds to baseline fluctuations, where the rate of change in resistance ($\frac{dR}{dt}$) remains near zero. Stage ② is initiated when $\frac{dR}{dt}$ exhibits a distinct shift from near zero to either a positive or negative value, marking the onset of the sensor response. During this stage, the system operates in a signal-tracking state. The baseline resistance R_0 is defined as the mean resistance over the 10 s preceding the response onset, and is updated for each response cycle to avoid feature extraction errors caused by baseline drift. As $\frac{dR}{dt}$ returns toward zero from either direction or reverses sign, the sensor progresses through the response stabilization (Stage ③) and recovery phases (Stage ④). When the system detects that it is in Stage ③, or transitions directly from Stage ② to Stage ④, R_g is calculated as the mean resistance over the 5 s preceding the stage transition, followed by derivation of the sensor response (a), where a response cycle is only identified if the response exceeds a predefined threshold, in order to avoid noise interference or false positives. Continuous monitoring of $\frac{dR}{dt}$ sign transitions ensures that the system effectively executes the full baseline–response–recovery cycle in a fully automated, real-time manner, while partially mitigating the effects of baseline drift.

3.2. Gaussian process regression interpolation

Gaussian Process Regression (GPR) was employed as the core regression model due to its strong capability in modeling nonlinear relationships and providing probabilistic predictions with uncertainty quantification. Unlike deterministic regression models, GPR defines a distribution over functions, enabling simultaneous estimation of both the predictive mean and associated confidence intervals. This property is particularly advantageous for gas concentration estimation under noisy and complex sensor response conditions. In addition, GPR performs well on small-to-moderate sized datasets, which matches the characteristics of the experimental data in this study, while maintaining strong generalization capability without requiring extensive parameter tuning. The model assumes that function values follow a multivariate Gaussian distribution, fully determined by a mean function $m(x)$ and a covariance function $k(x, x')$, which facilitates robust data augmentation and uncertainty-aware modeling in downstream prediction tasks.

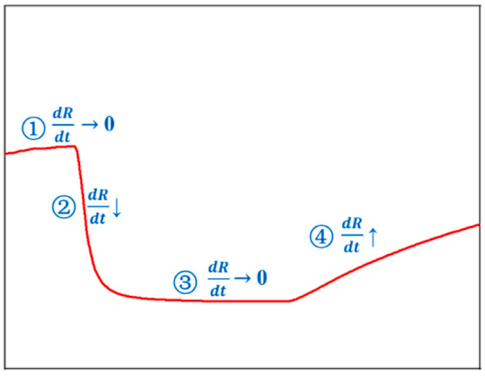


Fig. 4. Real-time feature extraction guided by the sensor signal derivative. The system automatically segments the sensor response into four stages (baseline, response onset, stabilization, recovery) based on $\frac{dR}{dt}$, enabling fully automated identification of R_0, R_g, and extraction of the steady-state response feature (a) in real time.

For a training set $D = \{(x_i, y_i)\}_{i=1}^{n}$, where the observations satisfy $y = f(x) + \epsilon, here \epsilon \sim N(0, \sigma_n^2)$ represents observational noise, the joint distribution of the training points and test points X_* is given by:

$$\begin{bmatrix} y \\ f_* \end{bmatrix} \sim N\left(0, \begin{bmatrix} K(X,X) + \sigma_n^2 I & K(X,X_*) \\ K(X_*,X) & K(X_*,X_*) \end{bmatrix}\right) \tag{3}$$

where $K(X,X)$ denotes the covariance matrix of the training set itself. According to the properties of conditional distributions, the posterior distribution of the test point f_* can be derived as:

$$f_* | X, y, X_* \sim N(\bar{f}_*, \mathrm{cov}(f_*)) \tag{4}$$

where the posterior mean $\bar{f}_*$ and covariance $\mathrm{cov}(f_*)$ are given by:

$$\bar{f}_* = K(X_*,X)\left[K(X,X) + \sigma_n^2 I\right]^{-1} y \tag{5}$$

$$\mathrm{cov}(f_*) = K(X_*,X_*) - K(X_*,X)\left[K(X,X) + \sigma_n^2 I\right]^{-1} K(X,X_*) \tag{6}$$

The posterior mean $\bar{f}_*$ serves as the predicted value, while the posterior covariance provides an estimate of predictive uncertainty. This property was leveraged to sample within high-confidence regions of the input space, generating probabilistically guaranteed synthetic samples and producing a high-quality augmented dataset. In this study, GPR was employed for data augmentation to mitigate the limitations of small sample training. A combination of radial basis function (RBF) and white noise kernel (White Kernel) was used, with the length scale and noise level boundaries set to (10^{-2}, 10^{2}) and (10^{-5}, 1e1), respectively.

To mitigate overfitting due to the limited dataset size and to enhance feature learning, data augmentation was applied exclusively during the pretraining stage of the contractive autoencoder (CAE). This augmented dataset was used to initialize and improve the feature extraction capability of the CAE, while avoiding potential bias in subsequent model training [32–34]. During the formal training phase, the CAE weights were frozen, and only the original measured dataset was used to train the DSN. This strategy increases feature robustness and data diversity without compromising the reliability of model evaluation.

3.3. CA-DSN model

The CA-DSN model was employed for concentration prediction. By integrating the robust feature extraction capability of the CAE with the stabilized training mechanism of the DSN, prediction accuracy and generalization performance were substantially enhanced in complex gas environments.

3.3.1. CAE

The encoder of the CAE was utilized as the feature extractor. In addition to the standard reconstruction loss of a conventional autoencoder, the CAE incorporates an additional regularization term to enhance feature robustness. The architecture of the model is illustrated in Fig. 5.

The loss function of the contractive autoencoder, $J_{CAE}(\theta)$, is defined as follows:

$$J_{CAE}(\theta) = \sum_{x \in D_n}\left[L(x, g(f(x))) + \lambda \left\| J_f(x) \right\|_F^2\right] \tag{7}$$

where $L(x, g(f(x)))$ denotes the mean squared reconstruction loss between the input feature x and the reconstructed output $g(f(x))$. $\|J_f(x)\|_F^2$ represents the squared Frobenius norm of the Jacobian matrix of the encoder (f); this term penalizes the sensitivity of the encoder output to input perturbations, enforcing the model to learn a "contractive" feature space that is insensitive to small variations in the input. The hyperparameter λ controls the strength of this regularization. Through this

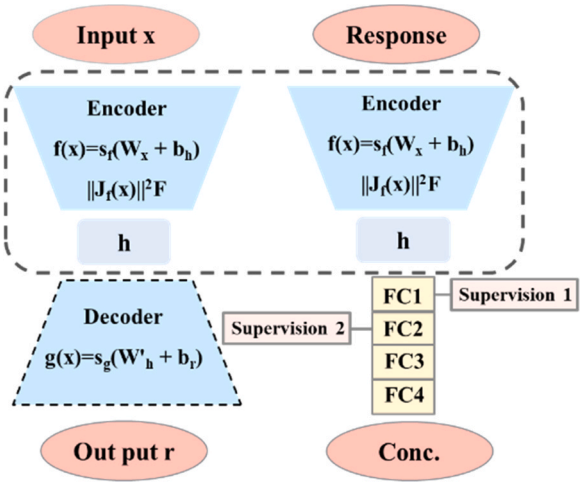


Fig. 5. Architecture of the Contractive Autoencoder and CA-DSN model. The CAE encoder maps the input sensor response x to a latent representation h, and the decoder reconstructs r. Contractive loss ($\left\|J_f(x)\right\|_F^2$) ensures feature robustness. Encoded features are fed into the DSN.

mechanism, the CAE effectively suppresses interferences and extracts robust features that capture the intrinsic properties of the gases, thereby enhancing model stability under complex environmental conditions.

3.3.2. DSN

To mitigate the vanishing gradient problem in deep networks and to guide intermediate layers toward learning more discriminative features, a deep supervision mechanism was applied to the encoder's intermediate layers, as illustrated in Fig. 6.

Specifically, in addition to the final main output layer, two auxiliary supervision branches (Supervision 1 and Supervision 2) were incorporated into two intermediate hidden layers of the network. Each branch consists of a fully connected layer performing the same concentration

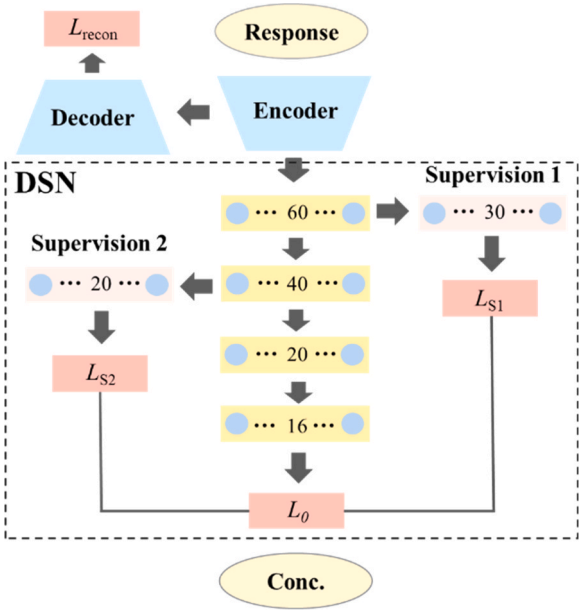


Fig. 6. Structure and training of the CA-DSN model. Robust features extracted by the CAE are processed through a deep network with two auxiliary supervision branches (Supervisions 1 and 2 at intermediate layers) plus the main output. This deep supervision mechanism accelerates convergence, mitigates vanishing gradients, and enforces hierarchical discriminative feature learning. Total loss: $L_{total} = L_0 + \alpha_1 L_{S_1} + \alpha_2 L_{S_2}$.

regression task as the main output. The total loss function (L_{total}) of the model is defined as the weighted sum of the main output loss and the auxiliary losses:

$$L_{total} = L_0 + \alpha_1 L_{S_1} + \alpha_2 L_{S_2} \tag{8}$$

where L_0 is the regression loss (MSE) of the main output layer, L_{S_1} and L_{S_2} are the regression losses of the auxiliary branches, α_1 and α_2 are hyperparameters used to balance the weights of each loss term.

The incorporation of DSN offers two key advantages. First, the auxiliary losses deliver stronger gradient signals to shallow layers during backpropagation, thereby accelerating convergence and enhancing training stability. Second, intermediate layers are encouraged to develop predictive capabilities, promoting the learning of hierarchical discriminative features from low-level to high-level representations.

3.4. PSO-based parameter optimization

Particle Swarm Optimization (PSO) [35,36] is a global optimization algorithm inspired by the collective behaviors of bird flocks and fish schools. The search for the optimal objective function is conducted through cooperative interactions among particles within the solution space. Each particle adjusts its search direction and step size based on its own best experience and the collective knowledge of the swarm, enabling effective tuning of model parameters.

In this study, PSO was adopted for hyperparameter optimization for several reasons. Compared with traditional grid search, PSO can efficiently explore high-dimensional parameter spaces with lower computational cost. Unlike gradient-based methods, it does not require a differentiable objective function, making it suitable for the non-convex optimization landscape of deep learning models. Moreover, compared with genetic algorithms, PSO generally achieves faster convergence while involving fewer tunable parameters, thereby reducing optimization overhead.

In a D-dimensional search space, assume the population $P = \{X_1, X_2, \ldots, X_M\}$ consists of M particles. The position vector of the i-th particle is represented as $X_i = (x_{i1}, x_{i2}, \ldots, x_{iD})$ and the velocity vector is represented as $V_i = (v_{i1}, v_{i2}, \ldots, v_{iD})$. Each particle records its individual historical best position as $P_i = (p_{i1}, p_{i2}, \ldots, p_{iD})$, while the global best position of the entire population is denoted as $G = (g_1, g_2, \ldots, g_D)$.

At each iteration, the particle updates its velocity and position according to the following formulas:

$$V_i(t+1) = \omega \bullet V_i(t) + c_1 r_1 \bullet (P_i(t) - X_i(t)) + c_2 r_2 \bullet (G(t) - X_i(t)) \tag{9}$$

$$X_i(t+1) = X_i(t) + V_i(t+1) \tag{10}$$

where ω is the inertia weight, which balances the global and local search capabilities; c_1 and c_2 are the learning factors that adjust the step size towards the individual best and global best positions, respectively; r_1 and r_2 are random numbers uniformly distributed in the range [0, 1]. The particles are iteratively updated, causing the swarm to gradually converge toward the global optimum of the objective function. The PSO process and corresponding implementation flowchart are shown in Fig. 7.

In this study, PSO was employed to automate the tuning of key hyperparameters in the CA-DSN model, with validation set performance serving as the fitness function. Through its collaborative search mechanism, the algorithm efficiently identified the optimal parameter combination, enhancing predictive performance and eliminating the need for manual trial-and-error tuning. The optimized parameters were as follows: batch size of 32, learning rate of 0.001, Adam optimizer, ReLU activation function, mean squared error (MSE) loss function, and supervision weight of alpha = 0.3.

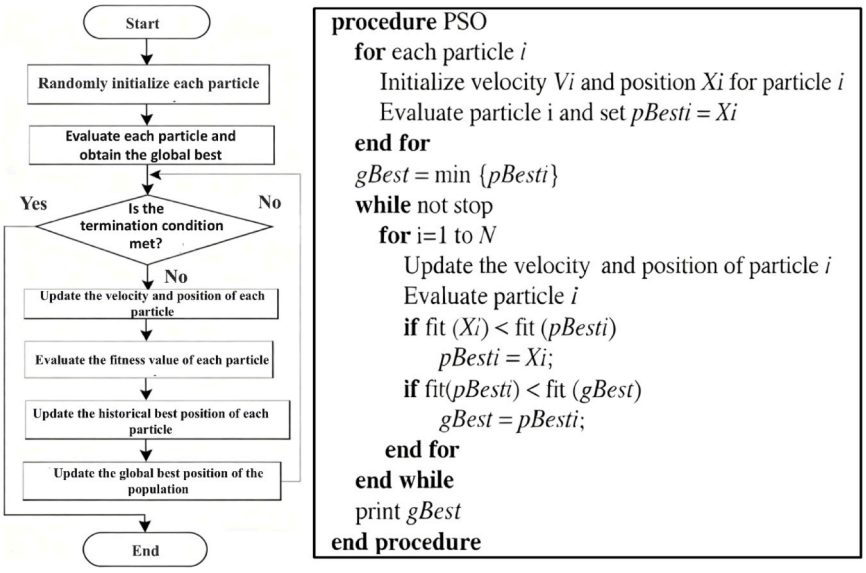


Fig. 7. Flowchart of the Particle Swarm Optimization (PSO) algorithm for CA-DSN hyperparameter tuning. (a) Overall iterative optimization process, showing particle initialization, evaluation, and update steps. **(b)** Pseudocode detailing velocity and position updates, and tracking of individual best (*pBest*) and global best (*gBest*) positions. The swarm efficiently identifies the optimal hyperparameter set by minimizing validation-set error as the fitness function.

3.5. End-to-end system workflow

The system is designed to achieve automated end-to-end operation, encompassing signal perception, feature extraction, concentration prediction, and model self-updating. Its core functionality is enabled by two main mechanisms: real-time response determination and intelligent calibration decision-making. The system's end-to-end processing workflow is illustrated in Fig. 8.

First, the system continuously monitors the raw resistance signals $R(t)$ output by the gas sensor array and computes their first derivative $\frac{dR}{dt}$ in real-time. The response phase of the sensor is determined based on the variation in $\frac{dR}{dt}$. During this period, the system automatically calculates R_g and computes the ratio of R_g to R_0 (i.e., the response degree a), which is then extracted as a valid steady-state feature vector for use in subsequent recognition or training processes.

During recognition, the extracted response features are fed into the trained CA-DSN model, which outputs the predicted concentrations of the mixed-gas components, thereby completing the real-time detection process in a fully automated end-to-end manner.

During model training and calibration, newly collected response data and their predicted results are temporarily stored in a buffer. When the accumulated data reaches a predefined threshold ($N_1 > n$), the system calculates the MAE of the predictions. Based on this error, a differentiated update strategy is applied: if the MAE falls below a preset threshold, lightweight parameter fine-tuning is performed. If the MAE exceeds the threshold, indicating a significant shift in data distribution or the emergence of new concentration scenarios, the new data are incorporated into the dataset and the model is retrained. To prevent error accumulation due to prolonged intervals without full retraining, accumulated samples exceeding a secondary threshold ($N_2 > m$) are forcibly added to the dataset, triggering preventive retraining, ensuring long-term reliability.

This workflow creates a closed-loop system that integrates real-time perception, intelligent decision-making, accurate prediction, and adaptive updating, effectively addressing the core challenges of real-time performance and long-term stability in electronic nose systems.

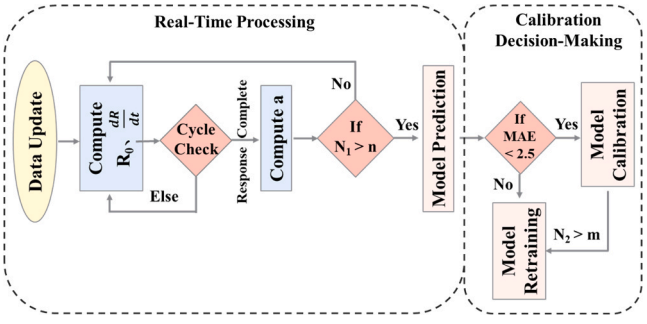


Fig. 8. End-to-end processing workflow of the intelligent electronic-nose system. Raw sensor signals are continuously updated, and the first derivative ($\frac{dR}{dt}$) and baseline resistance (R_0) are computed in real time to extract the steady-state response feature (a). During the model training process, the system evaluates whether the number of accumulated samples (N_1) exceeds the threshold (n) to determine if model prediction can proceed. Based on the predicted MAE, the intelligent calibration module either performs lightweight parameter refinement or triggers full model retraining, enabling fully automated, real-time maintenance and long-term robustness.

4. Results and discussion

The overall performance of the proposed real-time mixed-gas detection method, based on the CA-DSN model, was evaluated in this section. Experimental results are analyzed from three key perspectives: predictive performance, efficiency of the calibration strategy, and system performance under actual deployment conditions.

4.1. Data normalization

To eliminate the impact of varying dimensions and baselines in the raw sensor signals on model training, this study applied min-max normalization to all input data. This method linearly transforms the raw data to the range [0, 1], as shown by the following formula:

$$X_{norm} = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \tag{11}$$

where X represents the raw sensor response, $X_{\min}$ and $X_{\max}$ are the minimum and maximum values of the sensor response in the training dataset, and X_{norm} is the normalized output. By standardizing the numerical scale of each sensor, equal contribution from all sensors is

ensured during model training, preventing sensors with higher response amplitudes from disproportionately influencing the learning process.

4.2. Evaluation metrics

To assess the model's performance on mixed-gas concentration prediction, this study used Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE) as evaluation metrics. The mathematical expressions for these metrics are as follows:

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i| \tag{12}$$

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \tag{13}$$

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \tag{14}$$

MAE provides an intuitive measure of average prediction error, with smaller values indicating higher accuracy. MSE amplifies larger deviations by squaring them, increasing sensitivity to outliers and severe prediction errors. RMSE, while similarly sensitive to large errors, restores the original data scale through the square root operation, facilitating comparison with MAE while retaining MSE's sensitivity. The relative proximity of RMSE and MAE reflects the error distribution: similar values indicate relatively uniform errors, whereas a substantially larger RMSE suggests high variance in errors and lower prediction stability.

4.3. Model performance

The performance of the proposed CA-DSN model was compared with several baseline models, including the deep supervision network (DSN), long short-term memory supervised network (LSTMSN), convolutional supervised network (CSN), and Transformer. To ensure a fair comparison, all models were configured to use comparable numbers of neurons and hidden layers. Their hyperparameters were optimized to achieve the best performance using PSO.

Model performance was assessed using 10-fold cross-validation to minimize bias from a single data split. The dataset was randomly partitioned into 10 approximately equal, non-overlapping subsets. In each iteration, one subset was used as the test set, while the remaining nine subsets were used for training. Final performance metrics were obtained by averaging results across all folds.

4.3.1. VOCs

The CA-DSN model demonstrates significant advantages in predictive accuracy and robustness. As shown in Table 1, the model's MAE for C_2H_5OH, NH_3, and C_7H_8 are substantially lower than those of the baseline models, including the Transformer model, which exhibits higher MAEs of 1.61 ppm, 1.52 ppm, and 3.47 ppm for the respective gases, while incurring significantly higher computational costs (162k parameters, 5.18 M MACs). In contrast, the CA-DSN achieves MAEs of 1.20 ppm, 1.12 ppm, and 1.67 ppm for C_2H_5OH, NH_3, and C_7H_8, resulting in an overall MAE of 1.33 ppm with only 6.5k parameters and 174k MACs. Relative to the DSN, LSTMSN, CSN, and Transformer models, prediction accuracy is improved by approximately 36%, 44%, 49%, and 33%, respectively.

Furthermore, the CA-DSN model achieves low MSE and RMSE values of 3.46 and 1.86, respectively. Compared to the baseline models, MSE is reduced by 62%, 70%, 68%, and 56%, while RMSE is lowered by 39%, 44%, 43%, and 37%, demonstrating a significant performance improvement across all evaluated metrics.

These results confirm the effectiveness of the CAE in extracting robust features that are less sensitive to noise and emphasize the role of the DSN in enhancing generalization through stabilized training. Notably, the CA-DSN model maintained ppm-level prediction accuracy despite cross-interference arising from the similar sensor response patterns of C_2H_5OH, NH_3, and C_7H_8. Strong generalization was demonstrated across 120 samples (Fig. 9), underscoring the model's robustness and reliability. In the three binary mixture datasets composed of these three VOC gases, the CA-DSN also achieved the best results, with detailed comparisons shown in Tables S2–S4.

4.3.2. Thermal runaway characteristic gases

To evaluate the model's generalization under varying environmental conditions, the CA-DSN was tested on a mixed-gas dataset consisting of CO, H_2, and C_2H_2 collected at three different humidity levels. The quantitative results reported in the table demonstrate the model's strong predictive performance and robustness to humidity variations, extending the observations obtained from the VOC datasets.

In addition, to assess predictive performance under different dataset sizes and deployment-like scenarios, an incremental training scheme was adopted. For each humidity dataset, the model was fully retrained after every 10 newly acquired samples, resulting in six incremental training experiments per humidity condition, during which data acquisition and model training proceeded concurrently. The specific implementation details are provided in Section S1.

As shown in Table 2 and Fig. 10(b), which summarizes the average performance of all models across the three humidity datasets, the CA-DSN achieved the lowest overall mean absolute error (MAE = 0.67 ppm) and root mean square error (RMSE = 0.8), outperforming all

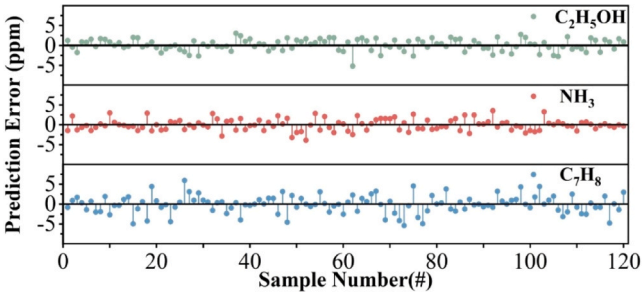


Fig. 9. Prediction errors of the CA-DSN model across the full concentration range. Data are aggregated from the test sets of all cross-validation folds. Detailed results are provided in Table S1.

Table 1
Model performance on the ternary VOCs dataset (C_2H_5OH, NH_3, C_7H_8).

Algorithms	Parameters	MACs	CO		H_2		C_2H_2	
			MAE (ppm)	RMSE	MAE (ppm)	RMSE	MAE (ppm)	RMSE
DSN	4.3k	124k	1.35	1.66	1.80	2.31	3.09	4.73
LSTMSN	50.0k	1.57 M	1.52	2.85	1.46	2.22	4.09	4.23
CSN	10.2 K	700k	2.67	3.41	1.94	2.13	3.2	3.91
Transformer	162k	5.18 M	1.61	2.00	1.52	1.78	3.47	4.24
CA-DSN	6.5k	174k	1.20	1.49	1.12	1.62	1.67	2.19

Table 2
Model performance under varying humidity conditions for ternary gas (H_2, CO, C_2H_8) prediction.

Algorithms	Dataset Size	Replications per Sample	CO		H_2		C_2H_2	
			MAE (ppm)	RMSE	MAE (ppm)	RMSE	MAE (ppm)	RMSE
DSN	60/60/60	1	1.48	1.81	0.46	0.62	0.59	0.71
LSTMSN			1.88	2.16	0.63	0.76	0.78	0.97
CSN			1.54	1.77	0.70	0.79	0.87	1.02
Transformer			1.47	1.76	0.42	0.56	0.66	0.88
CA-DSN			1.28	1.85	0.25	0.23	0.47	0.31

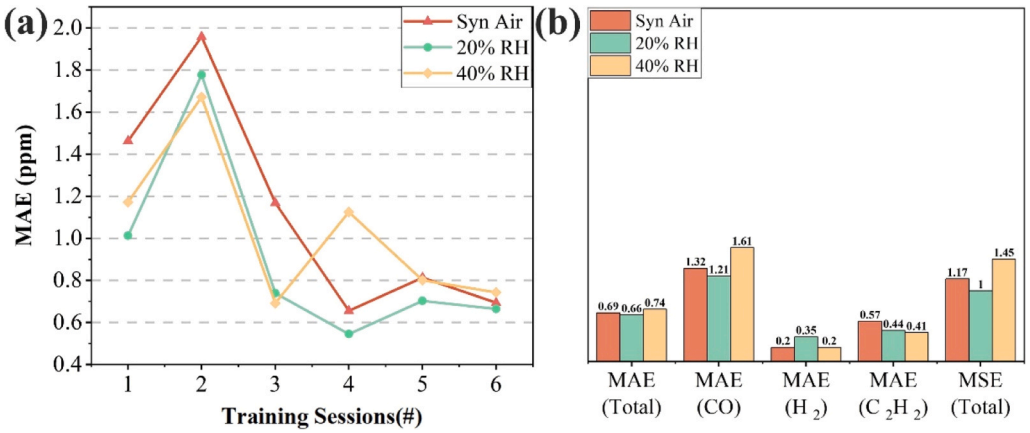


Fig. 10. Generalization performance of the CA-DSN model for H_2, CO, and C_2H_2 ternary mixtures under varying humidity conditions. (a) MAE trends over incremental training sessions in synthetic air, 20% RH, and 40% RH environments, illustrating robust convergence and stability under different humidity levels. **(b)** Comparison of final prediction performance across the three humidity conditions, confirming the model's high accuracy and strong robustness. Detailed results are provided in Tables S5,S6, and S7.

baseline models. A more detailed analysis reveals the model's advantages across several dimensions:

Compared to the second-best-performing Transformer model (MAE = 0.85 ppm), the CA-DSN achieved a reduction of approximately 21% in MAE. This further demonstrates that, even with significantly fewer model parameters and lower computational complexity, the CA-DSN still achieves higher accuracy. The MAE improvement is even more pronounced when compared to other baseline models (DSN, LSTMSN, CSN), with improvements ranging from 20% to 39%.

Furthermore, a component-wise analysis showed that the CA-DSN maintained high precision and stability under different humidity conditions. The CA-DSN's MAE for H_2 and C_2H_2 was as low as 0.25 ppm and 0.47 ppm, respectively, significantly outperforming other models in predicting these gases. Although the prediction error for CO was higher (MAE = 1.38 ppm), the model demonstrated considerable stability across different humidity levels, with the overall MSE controlled at 1.21. This stability, combined with the high precision for H_2 and C_2H_2, illustrates the CA-DSN's ability to effectively manage diverse gas response characteristics and potential cross-interference in complex mixtures, while maintaining strong adaptability to humidity variations.

Consistent with these observations, Fig. 10(a) shows that the model's performance converges rapidly, stabilizing after only four incremental learning runs. Such fast convergence with a limited number of samples suggests that the collected data possess sufficient information density, enabling the CA-DSN to learn robust and generalizable features even under data-constrained and dynamically evolving conditions.

Temperature is another important factor influencing sensor performance. In this study, the system's performance under varying temperature conditions was further investigated. Since the sensor temperature cannot be directly controlled, temperature variation was introduced by adjusting the heating voltage. Specifically, the model performance was compared at heating voltages of 5 V and 4 V under a fixed humidity condition of 20%. As shown in Fig. 11, although the sensor response

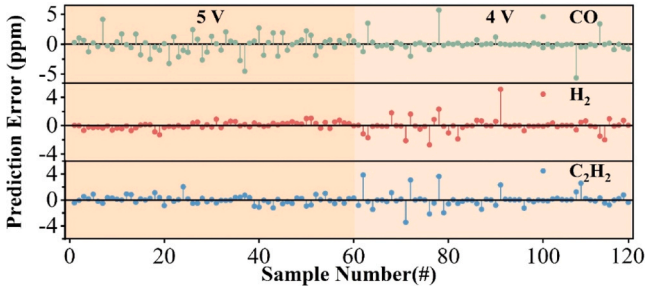


Fig. 11. Prediction errors of the CA-DSN model under different temperature conditions. All experimental settings, including the testing protocol, concentration combinations, and humidity, were kept identical except for the heating voltage. Among the 120 samples, the first 60 were collected at a heating voltage of 5 V, and the remaining 60 were collected at 4 V.

characteristics change with variations in heating voltage (as detailed in Tables S6 and S8), the signals still retain sufficient discriminative information for feature extraction. As a result, the overall recognition performance of the model is not significantly affected by temperature-induced variations.

4.3.3. Analyses

The comprehensive analysis of the model's performance across multiple datasets, including those collected under different humidity and temperature conditions, reveals the CA-DSN's robust generalization capabilities. As shown in the previous sections, the model not only demonstrates superior predictive accuracy and stability under different environmental conditions but also maintains its performance across various gas categories. The CA-DSN achieves a consistently lower error rate across both VOC and non-VOC gases, including CO, H_2, and C_2H_2, confirming its strong adaptability in real-world, multi-component gas

sensing applications.

Notably, the model's performance is influenced by both the dataset quality and sensor selectivity. In the case of the C_2H_5OH, NH_3, and C_7H_8 mixture, the gases produce similar sensor responses, leading to significant cross-interference and low feature distinguishability, which limits the model's ability to effectively differentiate between them. However, in the H_2, CO, and C_2H_2 mixture, the sensors exhibit better selectivity, resulting in more distinguishable responses and higher prediction accuracy.

This comparison highlights the importance of sensor selectivity in determining the model's performance. Despite the challenges posed by cross-sensitivity and dataset characteristics, the CA-DSN method effectively captures complex nonlinear relationships within the sensor signals, compensating for signal overlap. This significantly enhances the model's ability to handle complex mixtures, further demonstrating its potential for real-world deployment.

Moreover, by leveraging efficient feature learning and optimized model training, the CA-DSN can exploit the existing sensors' capabilities even in scenarios with limited sensor selectivity or dataset size. This flexibility reinforces the model's generalization and robustness, making it a promising solution for a wide range of multi-component gas sensing applications.

A comparison with representative recent studies is summarized in Table 3. The proposed CA-DSN exhibits favorable performance in terms of data efficiency, model compactness, and prediction accuracy. Ppm-level concentration estimation is achieved using a single sampling cycle and a relatively small dataset of 120 samples, whereas several existing methods require repeated sampling or substantially larger datasets. Meanwhile, CA-DSN adopts a compact network architecture with low parameter count and computational cost, making it more amenable to resource-constrained deployment scenarios. Despite its lightweight design, the model maintains competitive accuracy across both VOC mixtures and thermal runaway characteristic gases. Overall, the results in Table 3 indicate that CA-DSN provides an effective balance between sensing accuracy and computational efficiency compared with related approaches.

4.4. Model calibration

To evaluate the long-term effectiveness of the proposed real-time training and calibration strategy, a long-term operation scenario with environmental variations was explicitly simulated. Specifically, three thermal-runaway gas datasets collected under different humidity conditions were combined into a mixed-humidity dataset comprising 180 samples (synthetic air, 20% RH, and 40% RH). This setup emulates realistic humidity fluctuations encountered during prolonged system deployment. Experiments followed the baseline configuration described in Section 4.3.2, with the calibration decision module activated to enable autonomous model maintenance.

As shown in Fig. 11, the responses of individual sensors to H_2, CO, and C_2H_2 under the three humidity conditions reveal significant variations in both baseline and sensitivity due to humidity effects. Consequently, models trained on data from a single humidity condition lose predictive accuracy when applied under different environmental conditions. This experiment was designed to evaluate the system's capability to adapt to environmental variations, demonstrating whether the proposed adaptive calibration mechanism can maintain reliable prediction and operational stability under changing humidity levels, as further illustrated in Fig. 12.

The experimental results indicate that the model performance converges and stabilizes after approximately 40 samples. Once stabilized, newly acquired samples predominantly trigger the low-cost parameter refinement mechanism rather than full retraining. Although abrupt humidity variations affect sensor characteristics, induce feature drift, and temporarily degrade prediction accuracy—occasionally necessitating full model retraining—the model can rapidly adapt by learning generalized features from existing data augmented with only a small number of new samples. Notably, when the humidity level shifts to 40% RH, the model achieves accurate prediction after a single full retraining, after which it promptly reverts to the low-cost calibration mode.

As shown in Fig. 13, across a total of 18 calibration decision cycles, 10 low-cost parameter refinements were triggered (MAE $<$ 2.5 ppm), while only 8 full retraining events were required. Among these, three occurred during the model initialization stage, three were induced by abrupt humidity transitions, and one corresponded to a preventive retraining operation executed at the 12th session to mitigate potential error accumulation. These results demonstrate that the proposed system can sustain prediction performance with minimal computational overhead during long-term operation.

To quantitatively assess the trade-off between computational efficiency and prediction accuracy, the proposed intelligent calibration strategy was further compared with a conventional baseline in which full retraining is performed after each operation cycle. Computational cost analysis reveals that, although the calibration stage involves the same forward propagation as full retraining, only the backbone network parameters (accounting for approximately 72%) are updated during backpropagation. As a result, the per-sample computational cost of calibration is reduced to approximately 62% of that required for full retraining.

More importantly, the calibration process requires optimization using only the newly acquired 10 samples, whereas conventional retraining must process the entire accumulated dataset. As system operation progresses, this discrepancy in computational burden increases sharply. Experimental results show that by the 18th training

Table 3
Performance comparison between the proposed CA-DSN model and recent mixed-gas sensing models. To ensure a fair comparison, the reported parameter counts and multiply–accumulate operations (MACs), except for those of the proposed method, are taken directly from the corresponding references or estimated as the minimum theoretical values based on the model descriptions provided therein.

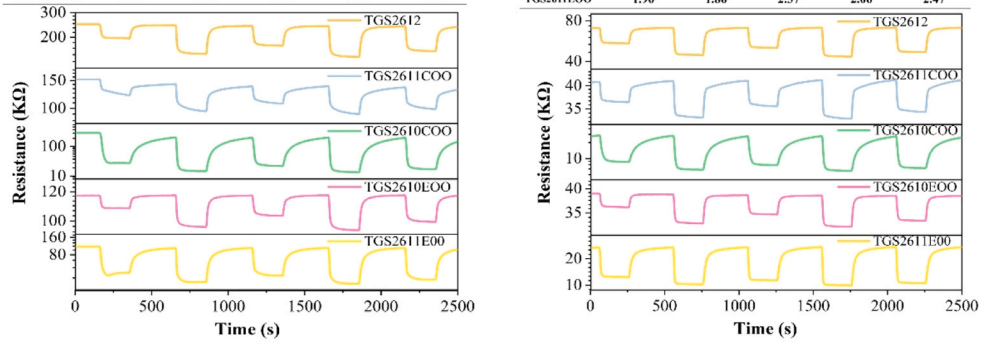
Gases ΔC(ppm)	Algorithms	Dataset Size	Replications per Sample	Parameters	MACs	Error Evaluation	Ref.
$H_2/CO/C_2H_4$ 30/20/20	CACNN-Mamba	650	10	> 6k	> 7 M	MAE (ppm) 1.4/2.4/2.5	[22]
H_2S/SO_2 5/5	SADDCN-HFF	-	-	77.6k	4.3 M	RMSE 4.62/0.35	[37]
CO/C_2H_4 5/5		525	21			RMSE 1.50/0.58	
CO/C_2H_4 5/5	VMD-LSTNet-Attention	525	21	> 160k	> 10 M	MAE (ppm) 1.33/2.02	[23]
CO/C_2H_4 5/5	RESHA-ALW	200	1	> 454k	> 6.5 M	RMSE 7.97/0.25	[38]
$C_2H_5OH/NH_3/C_7H_8$ 2.25/1.5/1.5	CA-DSN	120	1	6.5k	179.4k	MAE (ppm) 1.20/1.12/1.67	This work
$CO/H_2/C_2H_2$ 1.5/1.5/1.5		60/60/60	1			MAE (ppm) 1.28/0.25/0.47	

(a)

Unit : ppm	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5
CO	15.5	18.5	6.5	8	9.5
H_2	8	18.5	9.5	20	14
C_2H_2	11	12.5	14	20	18.5
TGS2612	1.27	1.26	1.61	1.39	1.66
TGS2611COO	1.21	1.27	1.49	1.29	1.50
TGS2610COO	7.28	10.46	13.87	9.06	14.82
TGS2610EOO	1.09	1.08	1.22	1.13	1.24
TGS2611EOO	2.61	3.13	3.86	3.00	4.10

(b)

Unit : ppm	Sample 1	Sample 2	Sample 2	Sample 2	Sample 2
CO	15.5	18.5	6.5	8	9.5
H_2	8	18.5	9.5	20	14
C_2H_2	11	12.5	14	20	18.5
TGS2612	1.34	1.30	1.59	1.40	1.64
TGS2611COO	1.13	1.12	1.23	1.16	1.25
TGS2610COO	4.52	4.51	6.94	5.14	7.19
TGS2610EOO	1.09	1.08	1.17	1.11	1.19
TGS2611EOO	1.96	1.86	2.37	2.06	2.47



(c)

Unit : ppm	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5
CO	15.5	18.5	6.5	8	9.5
H_2	8	18.5	9.5	20	14
C_2H_2	11	12.5	14	20	18.5
TGS2612	1.52	1.38	1.71	1.49	1.77
TGS2611COO	1.21	1.14	1.26	1.18	1.27
TGS2610COO	5.04	4.46	6.73	5.01	6.96
TGS2610EOO	1.14	1.10	1.21	1.13	1.22
TGS2611EOO	2.16	1.92	2.42	2.11	2.52

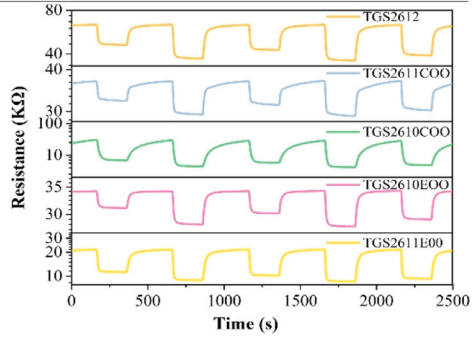


Fig. 12. Responses of sensors to mixed-gas under different humidity conditions. (a) synthetic air, **(b)** 20% RH, and **(c)** 40% RH, including the concentrations of each gas and the corresponding sensor response magnitudes.

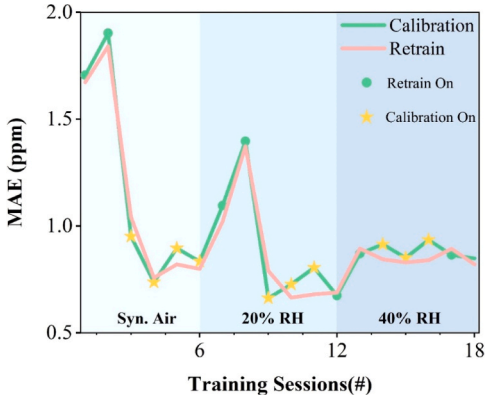


Fig. 13. Long-term evaluation of the adaptive calibration strategy under mixed-humidity conditions. Using a 180-sample dataset spanning synthetic air, 20% RH, and 40% RH, the system maintains stable prediction accuracy through autonomous switching between lightweight calibration and full retraining, achieving nearly 60% cumulative computational savings over 18 operation cycles.

session, the computational cost of the calibration strategy decreases to only 3.4% of that of full retraining. Over the complete 18-session operation, the self-calibration mechanism reduces cumulative computational consumption by nearly 60%, while maintaining essentially unchanged prediction accuracy.

4.5. *Actual prediction*

To assess the system's actual processing performance, the model for ternary gas mixtures from Section 4.3.1 was loaded., and predictions were performed on five sets of unprocessed raw sensor samples. The prediction results are presented in Fig. 14, demonstrating the system's capability to execute fully automated, end-to-end real-time detection on previously unseen data (Fig. 14).

The results displayed in Fig. 14 demonstrate that the system successfully achieves direct and accurate prediction of gas concentrations from raw sensor data, highlighting its robust end-to-end prediction capability. In the predictions for C_2H_5OH, C_7H_8 and NH_3, the system performed excellently, with average errors of only 0.95 ppm, 1.64 ppm and 1.88 ppm, respectively. Additionally, MSE of 3.48 and RMSE of 1.87 further confirm the reliability of the prediction results. These results provide supportive evidence that rapid and accurate gas identification can be achieved under deployment-like conditions, demonstrating the system's ability to handle raw, unprocessed sensor data in an end-to-end fashion.

No.	C_2H_5OH Actual	C_7H_8 Actual	NH_3 Actual	C_2H_5OH Predicted	C_7H_8 Predicted	NH_3 Predicted
1	13.5	1.5	24	13.09	5.43	20.66
2	9	4.5	3	8.98	6.40	5.20
3	45	24	13.5	45.61	23.33	12.97
4	15.75	18	15	14.76	18.33	13.27
5	40.5	7.5	9	37.76	6.14	7.39

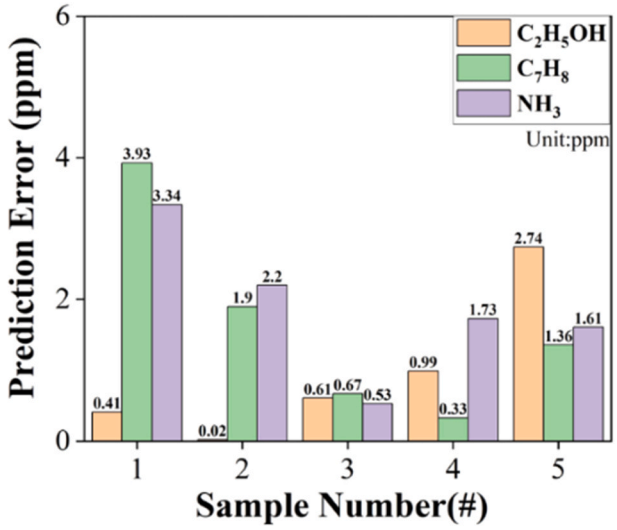


Fig. 14. Actual end-to-end prediction results of the deployed system on five previously unseen raw sensor samples. The system performs fully automated real-time detection, from signal acquisition to concentration prediction, achieving low average errors and demonstrating reliable generalization and automation in practical scenarios.

5. Discussion

The proposed CA-DSN-based electronic-nose system demonstrates reliable mixed-gas sensing performance across diverse gas categories, humidity conditions, and real-time scenarios. Its stable behavior under severe cross-interference and environmental perturbations suggests that the ultra-lightweight, automated end-to-end framework can effectively extract discriminative nonlinear features. The adaptive training and calibration strategy dynamically balances low-cost parameter refinement with full retraining, reducing cumulative computational overhead while maintaining prediction accuracy, thereby supporting long-term system operation.

It should be noted that this study aims to provide a feasible and practically valuable approach for mixed-gas concentration recognition. In real-world applications, constructing dedicated datasets that include both target and potential interfering gases remains necessary to ensure reliable performance. Furthermore, although the system exhibits good stability and robustness under varying environmental conditions, periodic calibration using standard gases is still recommended to mitigate the impact of irreversible sensor degradation during long-term deployment. Future work will focus on long-term monitoring of sensor performance evolution and conducting targeted investigations to further enhance the system's practical applicability.

6. Conclusions

This study presents an intelligent electronic-nose system capable of automated, end-to-end real-time gas detection. By combining dynamic signal–based feature extraction, a CA-DSN predictive model, and an adaptive calibration mechanism, the system operates in a "recognition while training" mode, achieving comparable predictive performance with approximately 60% reduced computational cost. Notably, only 40 initial labeled samples are required to establish a high-performance model. Experimental results confirm ppm-level accuracy and robust generalization across multiple mixed-gas systems and varying humidity conditions, while maintaining long-term stability and low maintenance. These findings suggest that the system exhibits adaptability to environmental variations and cross-system applicability, offering a feasible direction for reliable, low-power gas monitoring in practical gas sensing applications.

CRediT authorship contribution statement

Xin Guo: Writing – review & editing, Resources, Funding acquisition. **Qin-Kai Zou:** Writing – review & editing, Software, Methodology, Data curation, Conceptualization. **Yu-Ting Wang:** Supervision, Resources, Project administration. **Pu-Li Gan:** Writing – review & editing, Validation, Supervision. **Xuan-Zhen Lin:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.snb.2026.140277.

Data availability

Research Link Provided
CA-DSN (Figshare)

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